Week 7, Lecture 1. March 19, 2024

19-March-2024 TURING Machines.

Abstract model of Computation.

A TM is 7 tuple, $(Q, \Sigma, \Gamma, \delta, q_0, q_{accept}, q_{reject})$

Q set of states

Σ alphabet $\sqcup \notin \Sigma$

Γ tape alphabet including blank symbol $\sqcup$

$\delta : Q \times \Gamma \longrightarrow Q \times \Gamma \times \{L, R\}$

q_0

Configuration of a TM

$C_1 \to C_2 \to C_3 \to \cdots$

input, current state, tape position

tape content, current state, tape position

C_i $\boxed{1011q_7 01111}$ it represents

$\boxed{101101111}$ tape content,

q_7 is current state

C_{i+1} $\boxed{1011q_5 1111}$

$C_1 \to C_2$ C_1 yields C_2

A TM M accepts input w if a sequence of configurations $C_1, C_2, \ldots, C_k$ exist, where

1. C_1 is the start configuration of M on input $\underline{\underline{w}}$
2. each C_i yields C_{i+1}
3. C_k is an accepting configuration

The collection of strings a TM M accepts is called the "language of M" or language recognized by M. $L(M)$.

Turing-recognizable

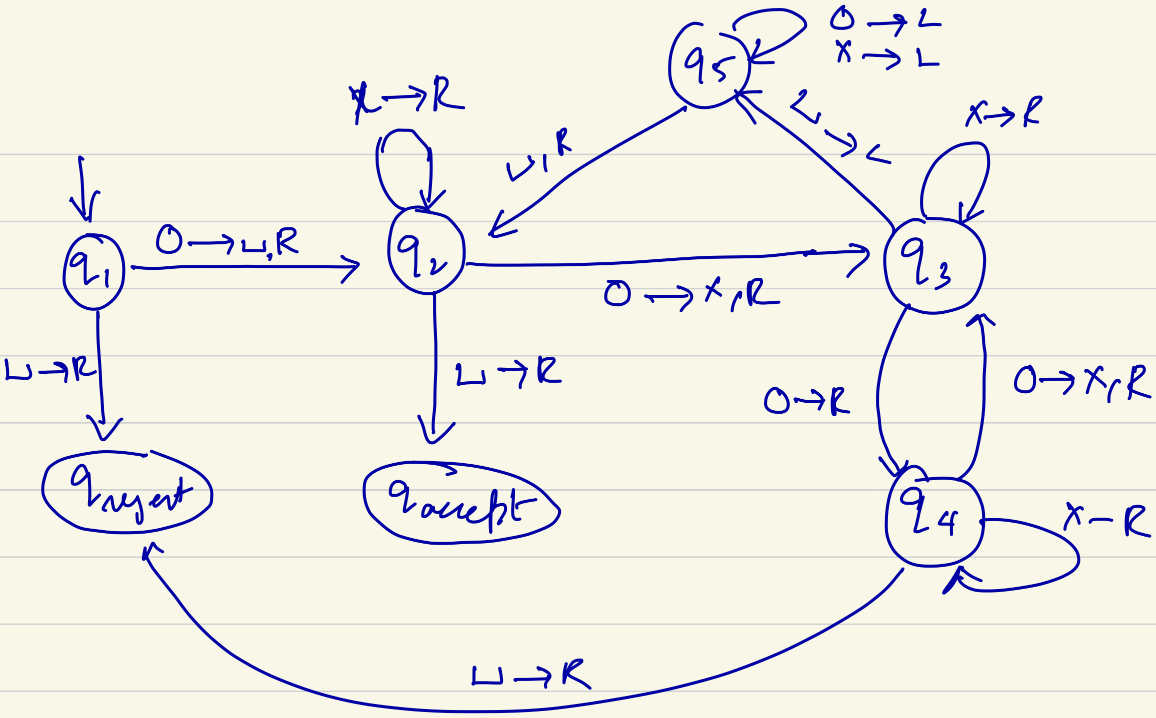
Turing-decidable.

$A = \{0^{2^n} \mid n \geq 0\}$ $\qquad \Sigma = \{0\}$

$A = \{0, 00, 0000, 00000000, \ldots\}$

M_A = " On input string w:

1. Sweep left to right across the tape, crossing off every other 0.
2. If in step 1 the tape contained a single 0, <u>accept</u>.
3. If in step 1 the tape contained more than a single 0 and the no. of 0's was odd, <u>reject</u>
4. Return the head to left-hand end of the tape.
5. Go to step 1."



$0 \longrightarrow \sqcup, R$